

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Subject: 392151 Chemistry I

Date: 15 August 2019

Year: 2019

Section: 15-18

Time: 8.00 a.m. – 10.00 a.m.

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 10 pages (including this page) and they are worth 70 points.

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

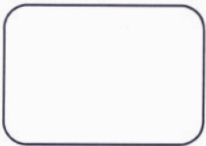
Periodic Table of the Elements

1	hydrogen 1 H 1.0079	2	helium 2 He 4.0026
3	lithium 3 Li 6.941	4	beryllium 4 Be 9.0122
5	sodium 11 Na 22.990	6	magnesium 12 Mg 24.305
7	potassium 19 K 39.098	8	calcium 20 Ca 40.078
9	rubidium 37 Rb 85.468	10	strontium 38 Sr 87.62
11	cesium 55 Cs 132.91	12	barium 56 Ba 137.33
13	francium 87 Fr [223]	14	radium 88 Ra [226]
15	scandium 21 Sc 44.956	16	titanium 22 Ti 47.867
17	yttrium 39 Y 88.906	18	zirconium 40 Zr 91.224
19	lanthanum 57 La 138.91	20	cerium 58 Ce 140.12
21	actinium 89 Ac [227]	22	thorium 90 Th 232.04
23	praseodymium 59 Pr 140.91	24	neodymium 60 Nd 144.24
25	europium 63 Eu 151.96	26	gadolinium 64 Gd 157.25
27	terbium 65 Tb 158.93	28	dysprosium 66 Dy 162.50
29	holmium 67 Ho 164.93	30	erbium 68 Er 167.26
31	ytterbium 70 Yb 173.04	32	lutetium 71 Lu 174.97
33	promethium 61 Pm [145]	34	samarium 62 Sm 150.36
35	europium 63 Eu 151.96	36	europium 63 Eu 151.96
37	gadolinium 64 Gd 157.25	38	gadolinium 64 Gd 157.25
39	terbium 65 Tb 158.93	40	terbium 65 Tb 158.93
41	holmium 67 Ho 164.93	42	holmium 67 Ho 164.93
43	ytterbium 70 Yb 173.04	44	ytterbium 70 Yb 173.04
45	lutetium 71 Lu 174.97	46	lutetium 71 Lu 174.97
47	beryllium 4 Be 9.0122	48	beryllium 4 Be 9.0122
49	lithium 3 Li 6.941	50	lithium 3 Li 6.941
51	sodium 11 Na 22.990	52	sodium 11 Na 22.990
53	potassium 19 K 39.098	54	potassium 19 K 39.098
55	rubidium 37 Rb 85.468	56	rubidium 37 Rb 85.468
57	cesium 55 Cs 132.91	58	cesium 55 Cs 132.91
59	francium 87 Fr [223]	60	francium 87 Fr [223]
61	scandium 21 Sc 44.956	62	scandium 21 Sc 44.956
63	yttrium 39 Y 88.906	64	yttrium 39 Y 88.906
65	lanthanum 57 La 138.91	66	lanthanum 57 La 138.91
67	actinium 89 Ac [227]	68	actinium 89 Ac [227]
69	praseodymium 59 Pr 140.91	70	praseodymium 59 Pr 140.91
71	neodymium 60 Nd 144.24	72	neodymium 60 Nd 144.24
73	europium 63 Eu 151.96	74	europium 63 Eu 151.96
75	gadolinium 64 Gd 157.25	76	gadolinium 64 Gd 157.25
77	terbium 65 Tb 158.93	78	terbium 65 Tb 158.93
79	holmium 67 Ho 164.93	80	holmium 67 Ho 164.93
81	ytterbium 70 Yb 173.04	82	ytterbium 70 Yb 173.04
83	lutetium 71 Lu 174.97	84	lutetium 71 Lu 174.97
85	beryllium 4 Be 9.0122	86	beryllium 4 Be 9.0122
87	lithium 3 Li 6.941	88	lithium 3 Li 6.941
89	sodium 11 Na 22.990	90	sodium 11 Na 22.990
91	potassium 19 K 39.098	92	potassium 19 K 39.098
93	rubidium 37 Rb 85.468	94	rubidium 37 Rb 85.468
95	cesium 55 Cs 132.91	96	cesium 55 Cs 132.91
97	francium 87 Fr [223]	98	francium 87 Fr [223]
99	scandium 21 Sc 44.956	100	scandium 21 Sc 44.956
101	yttrium 39 Y 88.906	102	yttrium 39 Y 88.906
103	lanthanum 57 La 138.91	104	lanthanum 57 La 138.91
105	actinium 89 Ac [227]	106	actinium 89 Ac [227]
107	praseodymium 59 Pr 140.91	108	praseodymium 59 Pr 140.91
109	neodymium 60 Nd 144.24	110	neodymium 60 Nd 144.24
111	europium 63 Eu 151.96	112	europium 63 Eu 151.96
113	gadolinium 64 Gd 157.25	114	gadolinium 64 Gd 157.25
115	terbium 65 Tb 158.93	116	terbium 65 Tb 158.93
117	holmium 67 Ho 164.93	118	holmium 67 Ho 164.93
119	ytterbium 70 Yb 173.04	120	ytterbium 70 Yb 173.04
121	lutetium 71 Lu 174.97	122	lutetium 71 Lu 174.97
123	beryllium 4 Be 9.0122	124	beryllium 4 Be 9.0122
125	lithium 3 Li 6.941	126	lithium 3 Li 6.941
127	sodium 11 Na 22.990	128	sodium 11 Na 22.990
129	potassium 19 K 39.098	130	potassium 19 K 39.098
131	rubidium 37 Rb 85.468	132	rubidium 37 Rb 85.468
133	cesium 55 Cs 132.91	134	cesium 55 Cs 132.91
135	francium 87 Fr [223]	136	francium 87 Fr [223]
137	scandium 21 Sc 44.956	138	scandium 21 Sc 44.956
139	yttrium 39 Y 88.906	140	yttrium 39 Y 88.906
141	lanthanum 57 La 138.91	142	lanthanum 57 La 138.91
143	actinium 89 Ac [227]	144	actinium 89 Ac [227]
145	praseodymium 59 Pr 140.91	146	praseodymium 59 Pr 140.91
147	neodymium 60 Nd 144.24	148	neodymium 60 Nd 144.24
149	europium 63 Eu 151.96	150	europium 63 Eu 151.96
151	gadolinium 64 Gd 157.25	152	gadolinium 64 Gd 157.25
153	terbium 65 Tb 158.93	154	terbium 65 Tb 158.93
155	holmium 67 Ho 164.93	156	holmium 67 Ho 164.93
157	ytterbium 70 Yb 173.04	158	ytterbium 70 Yb 173.04
159	lutetium 71 Lu 174.97	160	lutetium 71 Lu 174.97
161	beryllium 4 Be 9.0122	162	beryllium 4 Be 9.0122
163	lithium 3 Li 6.941	164	lithium 3 Li 6.941
165	sodium 11 Na 22.990	166	sodium 11 Na 22.990
167	potassium 19 K 39.098	168	potassium 19 K 39.098
169	rubidium 37 Rb 85.468	170	rubidium 37 Rb 85.468
171	cesium 55 Cs 132.91	172	cesium 55 Cs 132.91
173	francium 87 Fr [223]	174	francium 87 Fr [223]
175	scandium 21 Sc 44.956	176	scandium 21 Sc 44.956
177	yttrium 39 Y 88.906	178	yttrium 39 Y 88.906
179	lanthanum 57 La 138.91	180	lanthanum 57 La 138.91
181	actinium 89 Ac [227]	182	actinium 89 Ac [227]
183	praseodymium 59 Pr 140.91	184	praseodymium 59 Pr 140.91
185	neodymium 60 Nd 144.24	186	neodymium 60 Nd 144.24
187	europium 63 Eu 151.96	188	europium 63 Eu 151.96
189	gadolinium 64 Gd 157.25	190	gadolinium 64 Gd 157.25
191	terbium 65 Tb 158.93	192	terbium 65 Tb 158.93
193	holmium 67 Ho 164.93	194	holmium 67 Ho 164.93
195	ytterbium 70 Yb 173.04	196	ytterbium 70 Yb 173.04
197	lutetium 71 Lu 174.97	198	lutetium 71 Lu 174.97
199	beryllium 4 Be 9.0122	200	beryllium 4 Be 9.0122
201	lithium 3 Li 6.941	202	lithium 3 Li 6.941
203	sodium 11 Na 22.990	204	sodium 11 Na 22.990
205	potassium 19 K 39.098	206	potassium 19 K 39.098
207	rubidium 37 Rb 85.468	208	rubidium 37 Rb 85.468
209	cesium 55 Cs 132.91	210	cesium 55 Cs 132.91
211	francium 87 Fr [223]	212	francium 87 Fr [223]
213	scandium 21 Sc 44.956	214	scandium 21 Sc 44.956
215	yttrium 39 Y 88.906	216	yttrium 39 Y 88.906
217	lanthanum 57 La 138.91	218	lanthanum 57 La 138.91
219	actinium 89 Ac [227]	220	actinium 89 Ac [227]
221	praseodymium 59 Pr 140.91	222	praseodymium 59 Pr 140.91
223	neodymium 60 Nd 144.24	224	neodymium 60 Nd 144.24
225	europium 63 Eu 151.96	226	europium 63 Eu 151.96
227	gadolinium 64 Gd 157.25	228	gadolinium 64 Gd 157.25
229	terbium 65 Tb 158.93	230	terbium 65 Tb 158.93
231	holmium 67 Ho 164.93	232	holmium 67 Ho 164.93
233	ytterbium 70 Yb 173.04	234	ytterbium 70 Yb 173.04
235	lutetium 71 Lu 174.97	236	lutetium 71 Lu 174.97
237	beryllium 4 Be 9.0122	238	beryllium 4 Be 9.0122
239	lithium 3 Li 6.941	240	lithium 3 Li 6.941
241	sodium 11 Na 22.990	242	sodium 11 Na 22.990
243	potassium 19 K 39.098	244	potassium 19 K 39.098
245	rubidium 37 Rb 85.468	246	rubidium 37 Rb 85.468
247	cesium 55 Cs 132.91	248	cesium 55 Cs 132.91
249	francium 87 Fr [223]	250	francium 87 Fr [223]
251	scandium 21 Sc 44.956	252	scandium 21 Sc 44.956
253	yttrium 39 Y 88.906	254	yttrium 39 Y 88.906
255	lanthanum 57 La 138.91	256	lanthanum 57 La 138.91
257	actinium 89 Ac [227]	258	actinium 89 Ac [227]
259	praseodymium 59 Pr 140.91	260	praseodymium 59 Pr 140.91
261	neodymium 60 Nd 144.24	262	neodymium 60 Nd 144.24
263	europium 63 Eu 151.96	264	europium 63 Eu 151.96
265	gadolinium 64 Gd 157.25	266	gadolinium 64 Gd 157.25
267	terbium 65 Tb 158.93	268	terbium 65 Tb 158.93
269	holmium 67 Ho 164.93	270	holmium 67 Ho 164.93
271	ytterbium 70 Yb 173.04	272	ytterbium 70 Yb 173.04
273	lutetium 71 Lu 174.97	274	lutetium 71 Lu 174.97
275	beryllium 4 Be 9.0122	276	beryllium 4 Be 9.0122
277	lithium 3 Li 6.941	278	lithium 3 Li 6.941
279	sodium 11 Na 22.990	280	sodium 11 Na 22.990
281	potassium 19 K 39.098	282	potassium 19 K 39.098
283	rubidium 37 Rb 85.468	284	rubidium 37 Rb 85.468
285	cesium 55 Cs 132.91	286	cesium 55 Cs 132.91
287	francium 87 Fr [223]	288	francium 87 Fr [223]
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299	neodymium 60 Nd 144.24	300	neodymium 60 Nd 144.24
301	europium 63 Eu 151.96	302	europium 63 Eu 151.96
303	gadolinium 64 Gd 157.25	304	gadolinium 64 Gd 157.25
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315	lithium 3 Li 6.941	316	lithium 3 Li 6.941
317	sodium 11 Na 22.990	318	sodium 11 Na 22.990
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375	neodymium 60 Nd 144.24	376	neodymium 60 Nd 144.24
377	europium 63 Eu 151.96	378	europium 63 Eu 151.96
379	gadolinium 64 Gd 157.25	380	gadolinium 64 Gd 157.25
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Examination	Total score	Points
Part 1	12	
Part 2	58	
Total	70	

Answer sheet for Part 1

	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				



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Part 1: Multiple choices (12 points)

1. Which of the following choice could indicate that a chemical change has taken place?
 - (a) Combustion
 - (b) Distillation
 - (c) Evaporation
 - (d) Melting
2. An iron (Fe) bar has been left in the room where is humid since January 2019. Last month, a rust (Fe_2O_3) can be seen on the bar. Which choice is wrong about the rust formation?
 - (a) It is chemical change.
 - (b) It is physical change.
 - (c) It is reversible process.
 - (d) Both (b) and (c) are correct.
3. One property of most nonmetals is that they are
 - (a) Poor conductors of electric current
 - (b) Shiny
 - (c) Flattened when hit with a hammer
 - (d) Solid at room temperature
4. What is the difference between copper sulfate and copper sulfate crystals?
 - (a) Chemical property
 - (b) Physical property
 - (c) Intensive property
 - (d) Extensive property
5. When a metal and a nonmetal reacts, the _____ tends to lose electrons and the _____ tends to gain electrons.
 - (a) metal, metal
 - (b) nonmetal, nonmetal
 - (c) metal, nonmetal
 - (d) nonmetal, metal
6. Which technique is useful for refining crude oil?
 - (a) Sublimation
 - (b) Chromatography
 - (c) Fractional distillation
 - (d) Electrolysis

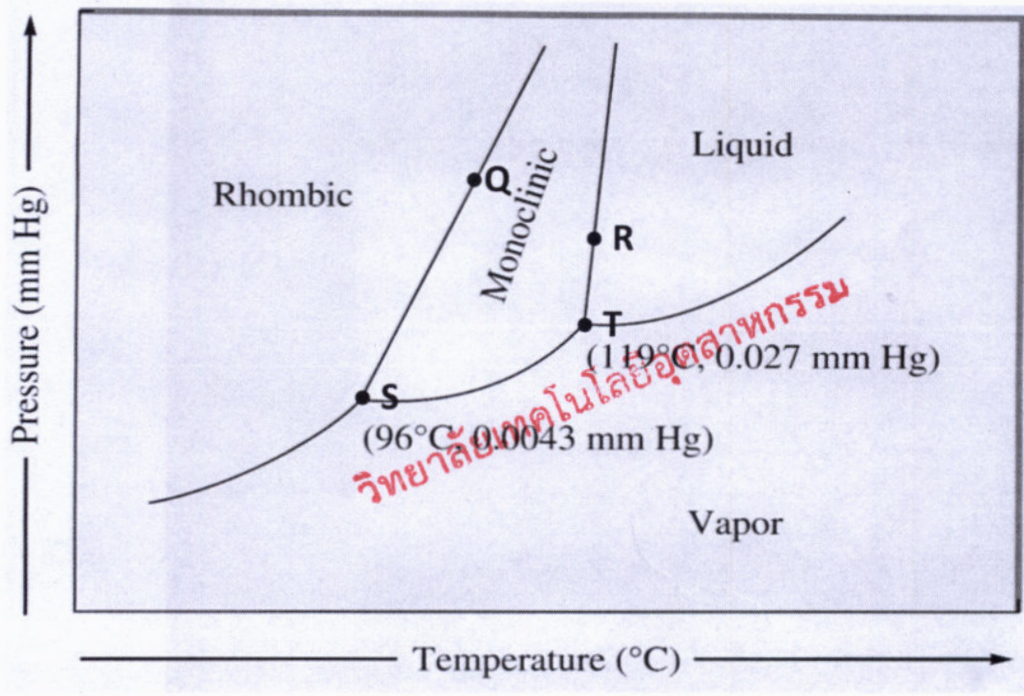
7. Which formula/name pair is incorrect?
- (a) $\text{Mn}(\text{NO}_2)_2$ manganese (II) nitrite
 - (b) $\text{Mg}(\text{NO}_3)_2$ magnesium nitrate
 - (c) $\text{Mn}(\text{NO}_3)_2$ manganese (II) nitrate
 - (d) Mg_3N_2 magnesium nitrite
8. Which of the following particles has the least mass?
- (a) Electron
 - (b) Neutron
 - (c) Proton
 - (d) Hydrogen nucleus
9. Thompson's model of the atom was:
- (a) planetary
 - (b) like plum pudding
 - (c) an empty space
 - (d) was similar to the model we have today.
10. Which statement is incorrect?
- (a) Rutherford found a dense nucleus located at the center of an atom.
 - (b) Dalton claimed an atom of the same element is not indifferent.
 - (c) Bohr's model can be applied for some elements well.
 - (d) Einstein used Planck's quantum theory to explain photoelectric effect.
11. The element with the electron configuration $[\text{Kr}]5s^24d^{10}5p^5$ is
- (a) representative element.
 - (b) transition metal.
 - (c) noble gas.
 - (d) an actinide element.
12. What is the total number of electrons in an Mg^{2+} ion?
- (a) 10
 - (b) 24
 - (c) 2
 - (d) 24

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Part 2 Answer the questions (58 points)

1. Matters and Changes

1.1 Sulfur in solid state can be either rhombic or monoclinic sulfur as displayed in the phase diagram of sulfur below. If there are 4 points of interest (Q, R, S and T), specify what you will observe at those points. If solid sulfur can be seen, you must indicate the form of sulfur. (6 points)



Q point:

R point:

S point:

T point:

If there is another point at 100°C and 0.02 mm Hg, indicate the state of sulfur including its detail.

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1.2 Copper covered in plastic has been used in electrical wiring. Explain and provide scientific detail why the electrical wire has been designed as mentioned. (4 points)

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2. Properties of Matters, Pure Substances and Solution

2.1 Classify the following materials as heterogeneous mixture, solutions, compounds, or elements. (4 points)

- a. concrete _____
- b. selenium _____
- c. baking powder _____
- d. ethanol _____
- f. condensed milk _____
- g. brass _____
- h. aluminum nitrate _____
- i. apple juice with pulp: _____

2.2 Name or write the formula of each of the following compounds (6 points)

- (a) $(\text{NH}_4)_3\text{PO}_4$: _____
- (b) N_2O_5 : _____
- (c) Mg_3N_2 : _____
- (d) KOH : _____
- (e) Aluminum sulfide: _____
- (f) $\text{Fe}_2(\text{SO}_4)_3$: _____
- (g) Calcium carbonate: _____
- (h) Chromium(III) nitrate: _____
- (i) $\text{Hg}(\text{NO}_3)_2$: _____
- (j) P_4O_6 : _____
- (l) Chromium (III) nitrate: _____
- (m) Phosphorus pentachloride: _____

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3. Separation of Matters, Energy and Changes

3.1 There is a mixture composed of iron filling and wet sand. Can you suggest any separation technique (s) in order to separate all the components of the mixture? Provide the detail of each separation technique to support your answer. (4 points)

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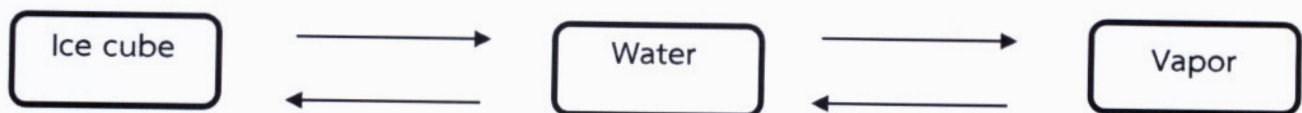
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3.2 Consider the diagram of change in states of water as shown below and write proper scientific terms above the arrows. (4 points)



3.3 Specify the following sentence whether it is right or wrong and correct the sentence if some content is not right. "Lighting fireworks is a reversible endothermic process since heat is released." (2 points)

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4. Atomic Structures, Fundamental Particles of Atoms

4.1 Choose words from the list as shown below to fill in the boxes. (6 points)

- | | |
|--------------------------------------|--|
| a. Orbital | ☐ Speed of light in vacuum |
| b. Isotone | ☐ Proton mass |
| c. Isotope | ☐ electron mass |
| d. Isobar | ☐ The sum of the number of protons and the number of neutrons. |
| e. Mass number | ☐ Having the same number of protons but different numbers of neutrons |
| f. Atomic number | ☐ Which subatomic particle has no charge? |
| g. Electron | |
| h. Neutron | |
| i. Proton | |
| j. $3 \times 10^8 \text{ m/s}$ | |
| k. $1.67 \times 10^{-27} \text{ kg}$ | |
| l. $9.1 \times 10^{-31} \text{ kg}$ | |

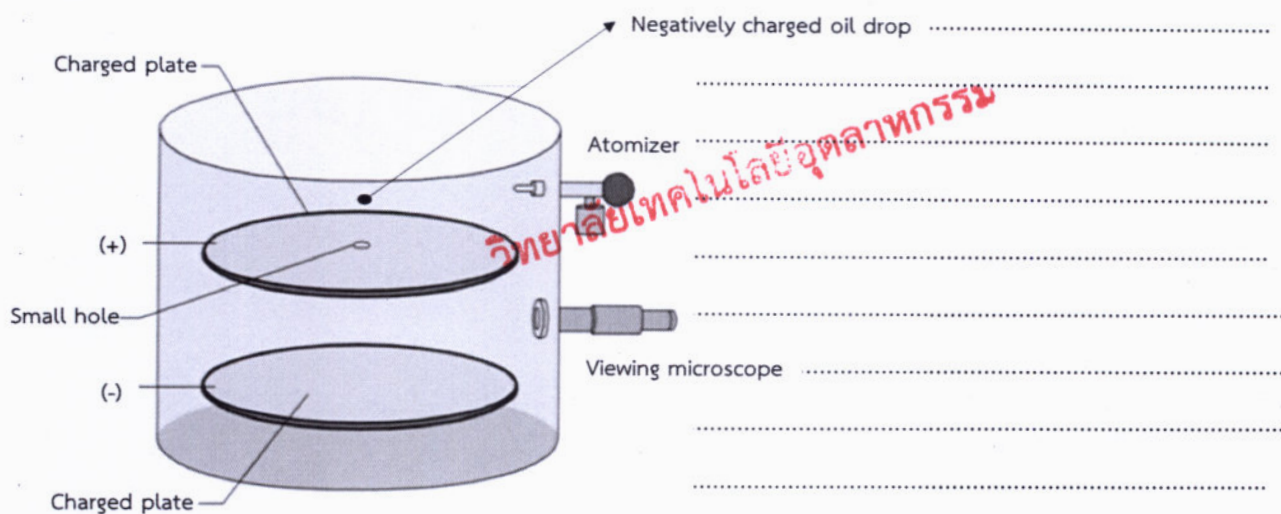
4.2 Use the periodic table to fill in the following chart and write the electron configurations for the following ions and elements. (4 points)

	No. of electrons	No. of neutrons	Nuclear symbol	Electron configuration (using the subshell model)	Electron configuration (using the shell model)
Mg					
Mn ²⁺					
Cu					
Kr					

5. Atomic Models

5.1 Based on JJ Thomson's atomic model, draw an atom of carbon. (3 points)

5.2 Oil drop experiment was designed to determine a weight and a charge of an electron. Explain how a negatively charged oil drop travels between 2 charged plates in the oil drop experimental setup (See the figure below) including scientific reasons. (3 points)

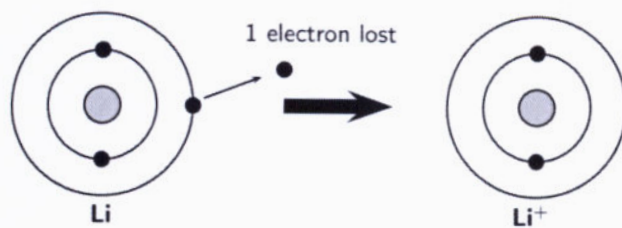


5.3 Calculate energy change and predict the color of the light emission, if an electron changes the orbit from $n=3$ to $n=2$. What region of electromagnetic spectrum is this radiation found? Hint: Use the information demonstrated in the bottom right corner wisely. (4 points) ($R_H = 2.12 \times 10^{-18} \text{ J}$, $c = 3 \times 10^8 \text{ m/s}$ and $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$.)

- **Violet:** 400 - 420 nm
- **Indigo:** 420 - 440 nm
- **Blue:** 440 - 490 nm
- **Green:** 490 - 570 nm
- **Yellow:** 570 - 585 nm
- **Orange:** 585 - 620 nm
- **Red:** 620 - 780 nm

6. Electron Configuration

6.1 Use the diagram for lithium as a guide and draw similar diagrams to show how each of the following ions is formed. (4 points)



Ca ²⁺	
O ²⁻	

6.2 Determine what group (1-18) and period of the elements, which are denoted by the following electron configurations. (4 points)

	group (1-18)	period
$1s^2 2s^2 2p^6 3s^2 3p^6$		
$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1$		
[Kr] $5s^2 4d^{10} 5p^1$		
2 8 18 2		

Assist. Prof. Dr. Nathjanan Jongkon and
Dr. Sunanta Chuayprakong